

Divyanshu Agarwal



Suwon-si, Korea

English, Hindi, Korean(A2)

EDUCATION

2024 IIT Delhi, B. Tech. in Computer Science and Engg. (GPA: 9.65/10)

- Exchange student at **University of Waterloo** (Fall 2022), ranked 1st among 1200+ sophomores
- Received **Endowment Merit Scholarship**(2023) & **IIT Delhi Merit Prize** in 5 semesters for academic excellence
- Awarded the **Summer Undergraduate Research Award** (2022) by the Industrial R&D Unit IIT Delhi
- **Teaching Assistant** for Intro to Comp. Science, conducted tutorials, designed and graded exams for **600+** students

ACADEMIC ACHIEVEMENTS

- Secured **AIR 3 / 1.2M** candidates in **JEE Main 2020**, India's largest engineering selection examination
- Achieved **AIR 97 / 150K+** candidates in **JEE Advanced 2020**, earning admission to **IIT Delhi (CSE)**
- Ranked **AIR 161** in the **KVPY Fellowship Exam 2019** by the Indian Institute of Science (IISc)
- Awarded the **NTSE Scholarship (Top 1,000 nationally)** in 2018, recognizing exceptional academic performance

WORK EXPERIENCE

Samsung Electronics (HQ), South Korea — *ML Engineer & Technical Lead* Sep 2024– ongoing
Visual Display Division – AI & Big Data Lab || C/C++, AI/ML, Git, Linux

- Leading **4** cross-office projects across **Delhi–Nanjing–Seoul**, coordinating **15+ engrs.** driving ML development
- Feature owner for **SmartThings TV IOT Sensors App** (C++), managing sensor capability APIs (huesync, sound, light), and defect resolution (~30/yr) across **300+** software models powering **200M+ users globally**
- Built the **Camera IoT** service with **30-fps** live-streaming & **on-device** object detection, serving **60M+ devices**
- Optimized **MobileNet-v1** for sound-event detection to reduce size **18.3→13.7 MB** via mixed-precision quantization
- Developed on-device recognition for **250+** landmarks for **Tizen XR** using **EfficientDet**, achieving **78.5%** accuracy
- Improved **KeyBERT** based Sentence Keyword Extraction by knowledge distillation, raising mAP from **0.438→0.485**

INTERNSHIPS

Samsung Electronics (HQ), South Korea — *Software Engineering Intern* Jun-Aug 2023

- Trained an **LSTM+Attention** model for exercise recognition using **1.3k UCF-101 clips** (7s each) and **MediaPipe**
- Boosted accuracy from **85%→92%** via frame interpolation and built a real-time app using OpenCV for pose feedback
- Refactored **Tizen's XR Framework**, enhancing **maintainability** and enabling pose, gesture & music recognition

HAL Research Group, University of Wisconsin — *GPU Accuracy Profiling* May-Aug 2022

- Tested **18+ microbenchmarks** (latency, bandwidth, atomics) to profile Gem5 against **AMD GCN/HBM GPUs**
- Identified ~**70%** discrepancy in **Cache/DRAM** behavior and calibrated latency parameters to cut down to **10%**

PROJECTS

NSCL : Neuro-Symbolic Concept Learner (Prof. Parag Singla) Aug-Dec 2023

- Built an end-to-end AI model to learn **20+ inductively generalisable concepts** from few shot demonstrations
- Generated a **50k+ sample PyBullet** dataset of simulations and achieved **12×** speedup through parallelization
- Implemented an LLM-driven symbolic module with **MCTS-based concept induction** and program synthesis
- Developed a neural-only baseline using a **diffusion model**, contributing to a co-authored **ICRA 2026** submission

CineBase : Database-Backed Web App Jan-April 2023

- Normalized the IMDb dataset of **106.2M (8.3GB → 5.2GB)**-records using ER modeling, reducing storage by **37%**
- Built optimized **PostgreSQL** pipelines (cleaning, joins, aggregates) to power recommendation across **10.6M** titles
- Cut search latency from **30s+** to **<2s** by adding **B-tree indexes**; and using **CTEs**, triggers, full-text search queries

Prophet Inequalities and Online Selection Algorithms (Prof. Ashish Chiplunkar) Aug '23 –Apr 2024

- Improved the SOTA guarantee for the **semi-online multi-selection** prophet algorithm from $1 - \frac{1}{2e}$ to $1 - o(1)$
- Constructed a novel **3-point** distribution proving an **upper bound of 0.90**, tightening limits for semi-online selection
- Used duality and stochastic dominance to express the optimal payoff for IID prophet inequality as a 2^{nd} order DE

CNN Classifier from Scratch for CIFAR-10 data-set Jan-Apr 2023

- Built a 3-layer CNN(**127k params; mixed 3×3/5×5 kernels**, dropouts), trained with Adam+CosineLR → **62.4%**
- Reimplemented using **PyTorch** with ~**220k params**, augmentation and GPU training → **71.8%** on 50k images

INTERESTS

I am a Xenophile that loves Word games and Philosophical Fiction. I also enjoy Running (5-10k) & Strength Training.